

Drug Delivery: Engineering and Principles

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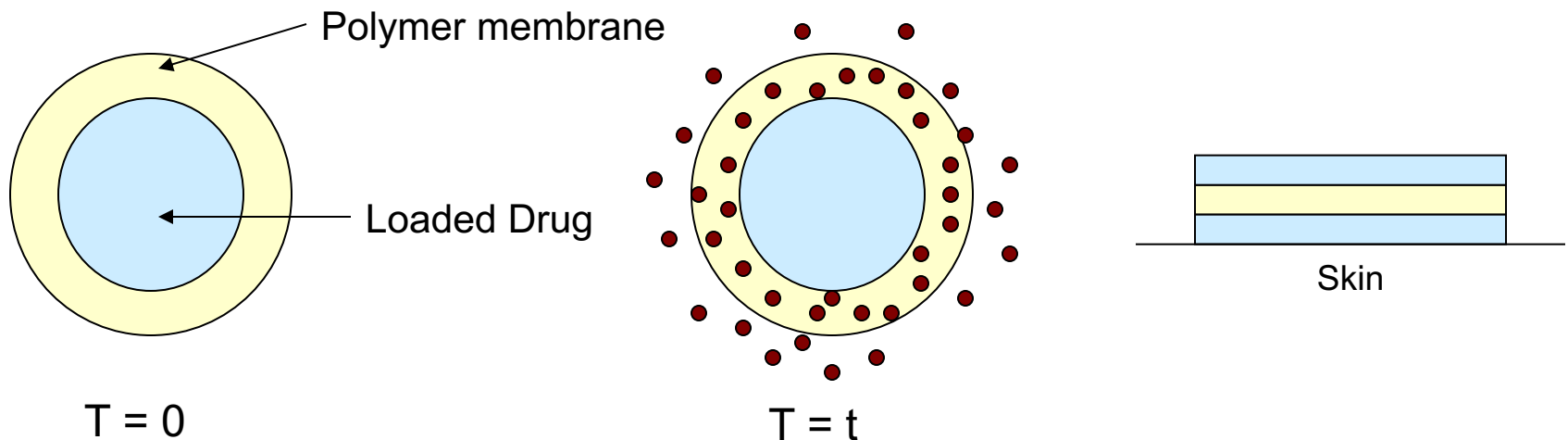
Bioengineering (BE)

Indian Institute of Science

Reservoir System

A supersaturated drug reservoir, surrounded by a non-degradable polymer membrane

- Could have planar configuration (reservoir between two membranes, e.g. Ocusert) or reservoir above a polymer membrane (e.g. skin patch)
- Could have cylindrical configuration (drug loaded in a tube like device, e.g. Norplant contraceptive device)
- Most commonly used polymers: Silicone elastomers, EVAc
- Could achieve nearly constant (zero-order) release rates



Reservoir Systems

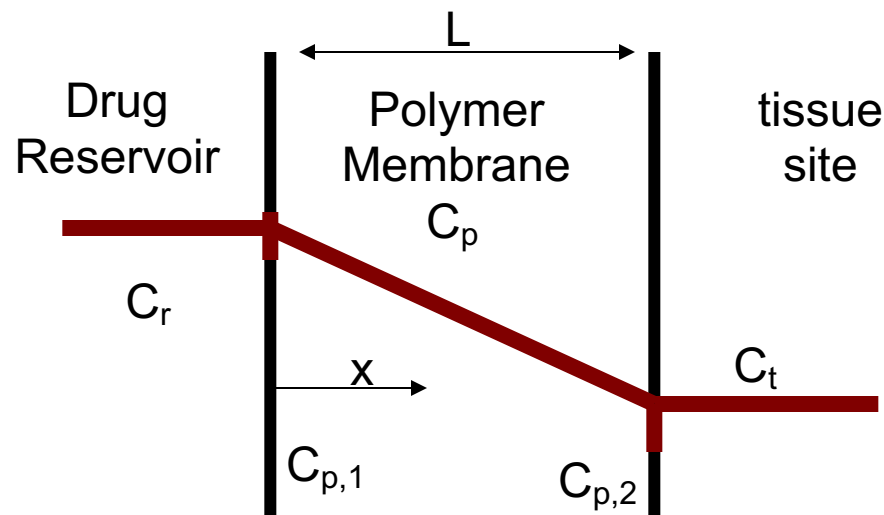
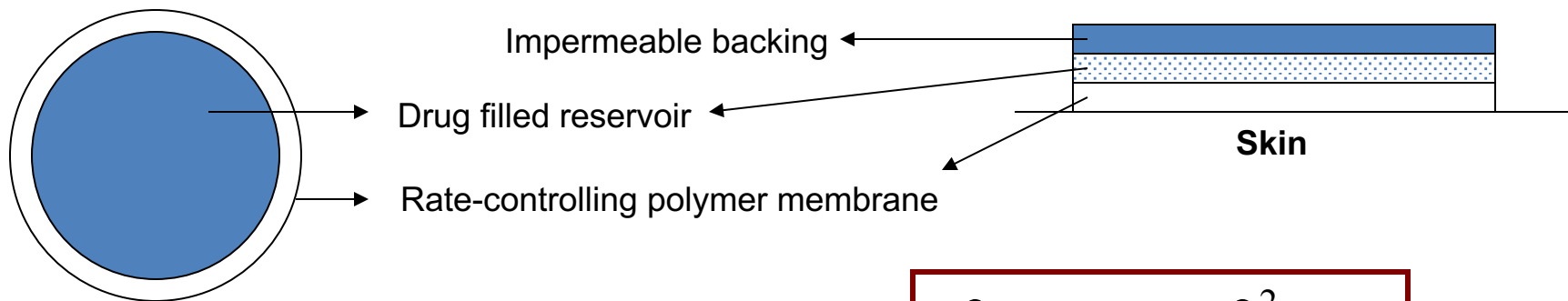
- **Advantages**

- Zero order (constant) release
- Easy to control kinetics by device design parameters

- **Disadvantages**

- Non-degradable, must be removed
- Impermeable to high molecular weight drugs (low porosities)
- Leaks can be dangerous (spilling of supersaturated drug)
- Cost of surgeries

Diffusion controlled delivery: Reservoir type system



$$\frac{\partial c_p}{\partial t} = D_{i:p} \frac{\partial^2 c_p}{\partial x^2} \quad \dots (1)$$

$$c_p = c_{p,0}; t = 0; 0 < x < L$$

$$c_p = c_{p,1}; t > 0; x = 0$$

$$c_p = c_{p,2}; t > 0; x = L$$

$C_{p,0}$ = initial concentration in membrane (uniform)

Perfect sink conditions

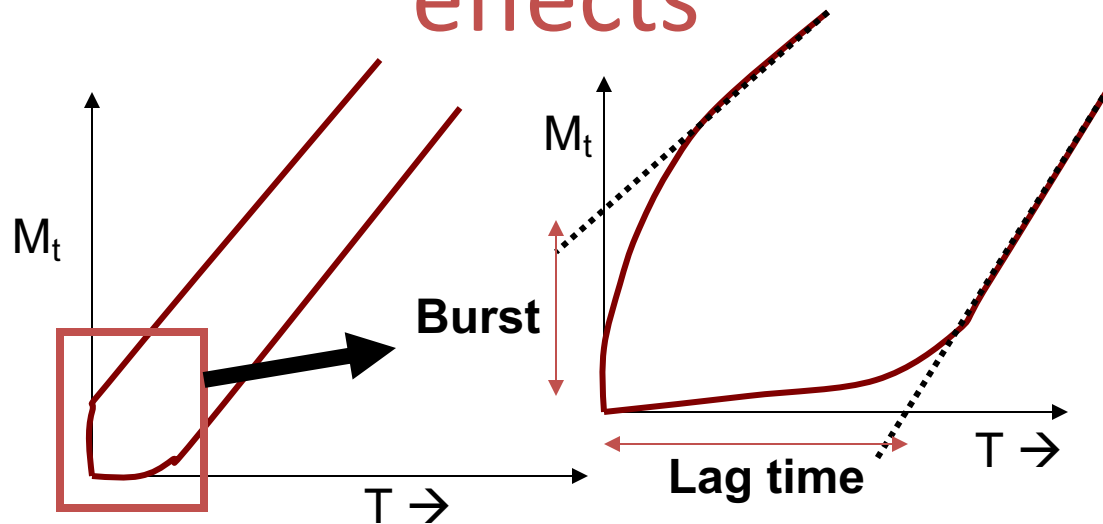
Release kinetics: Reservoir systems

- Membrane-reservoir

$$M(t) = \frac{ADKC_s}{l}t$$

- $M(t)$ is the amount of drug release at time t , A is the surface area of release, D is the drug diffusion coefficient through the polymer carrier, K is partition coefficient, C_s is the drug solubility in the polymer carrier, l is the thickness of the rate-controlling membrane, C_0 is the initial drug loading while t is the duration of release.

Reservoir systems: Lag and burst effects



- Due to the “time history” of the device
- Device used after storage → burst effect
 - Membrane is saturated with drug
 - Upon contact with tissue/aqueous medium, initial high level of drug release followed by steady state constant release
- Device used immediately after fabrication → lag effect
 - Membrane is devoid of any drug
 - Time for the drug to diffuse through the thickness of the membrane
 - Initial slow rate followed by steady state condition

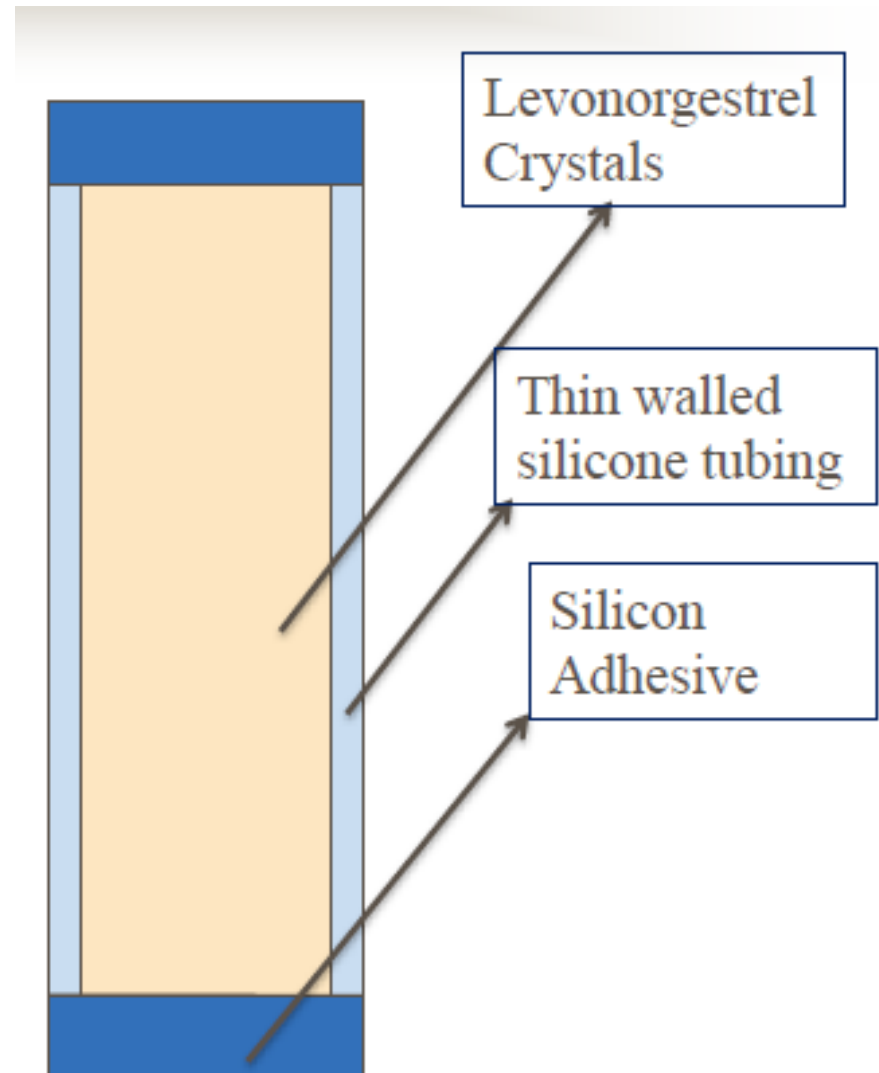
Example: Reservoir system

- Norplant Subdermal Implant:
 - Contraceptive implant
 - Six flexible closed capsules of Dimethylsiloxane / methylvinylsiloxane copolymer containing Levonorgestrel
 - Capsules are surgically implanted in the upper arm
 - Implant releases contraceptive drug continuously at the rate of $30\text{ }\mu\text{g/day}$ over a 5 year period
 - Capsules must be removed surgically
 - Upon removal patients promptly return to fertility

Norplant Contraceptive for Women

Polymer membrane
permeation-controlled
drug delivery system

The thickness of the
wall controls the
rate of the drug release



Reservoirs with Orifice

Nifedipine delivery orally for blocking calcium channels in hypertension

Procardia XL tablets

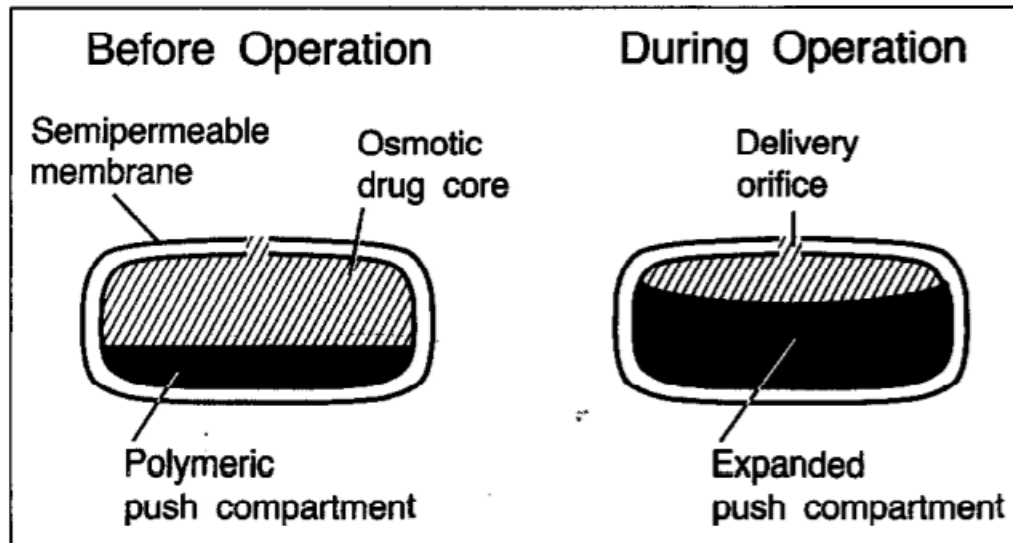
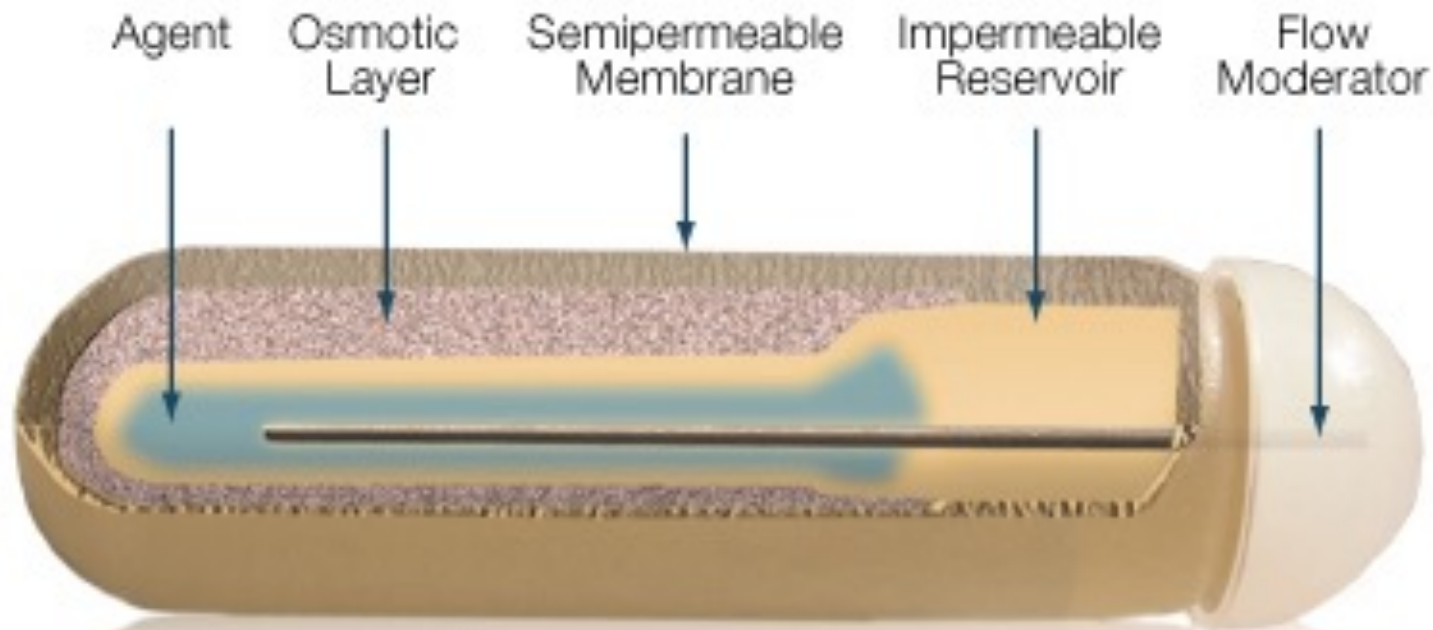


Figure 1. Cross section of the push-pull osmotic pump.

Half life of drug is 2h but this system results in sustained delivery for 24 hours. The device is then excreted in faeces

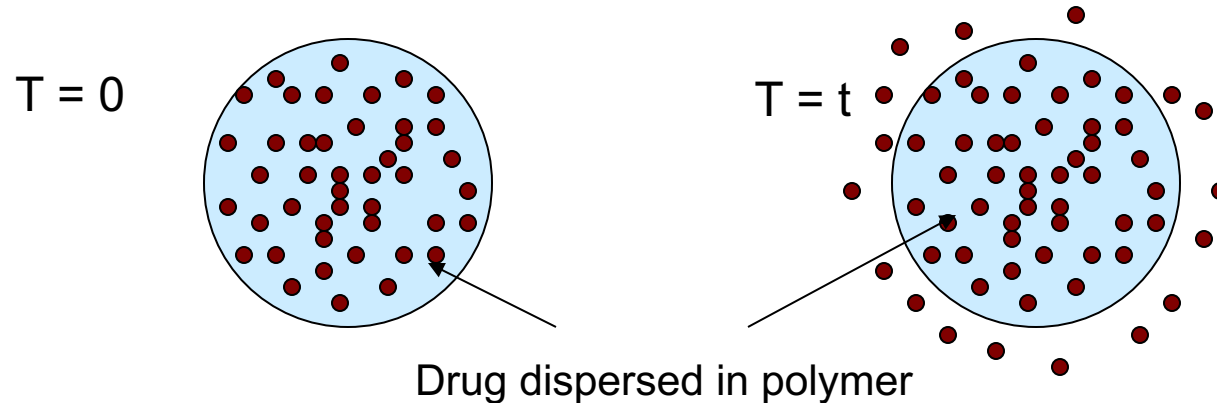


- Rate of delivery is controlled by the water permeability of the outer membrane
- Independent of the drug formulation
- Rates between 0.11 and 10 $\mu\text{L/hr}$
- Durations between 1 day and 6 weeks
- Flow modulator:
 - Minimize diffusion
 - Prevent large accidental spill



Mini osmotic pumps (Alzet Inc.)

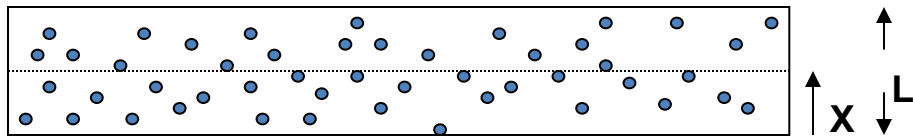
Non-erodible matrix systems



- **Advantages:**
 - Easy to fabricate
 - Less severe problem with leaks and cracks
 - Can be suitable for high molecular weight drugs
- **Disadvantages:**
 - Non-degradable, needs to be removed at end of delivery period
 - Release rate is not generally zero order

Dissolved drug, diffusion through polymer

- **Loading < solubility**
- Rate-limiting step is diffusion through polymer
- This can be made by simply soaking (absorption) the polymer matrix in a drug solution → conc. of drug solution = final conc. in matrix
- Therefore, the release can be modeled as a simple desorption problem



$$\frac{\partial c}{\partial t} = D_{i:p} \frac{\partial^2 c}{\partial x^2}$$

Initial Condition

$$c = c_0; t = 0; 0 < x < L$$

Boundary Conditions

$$c = c_{ext}; t > 0; x = 0, L$$

$$\frac{\partial c}{\partial x} = 0; t > 0; x = L/2$$

Dissolved drug, diffusion through polymer

- These equations are used to simplify the release kinetics description to early and late time approximations for a slab:
- For amount released:

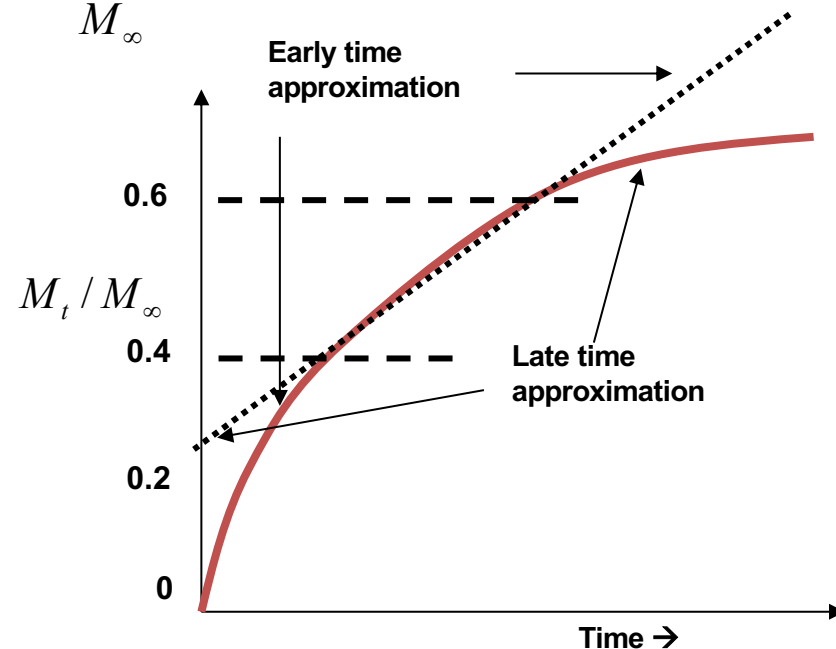
$$\text{EARLY TIME } \frac{M_t}{M_\infty} = 4\left(\frac{Dt}{\pi L^2}\right)^{1/2} \text{ for } 0 \leq \frac{M_t}{M_\infty} \leq 0.6$$

$$\text{LATE TIME } \frac{M_t}{M_\infty} = 1 - \frac{8}{\pi^2} \exp\left(-\frac{\pi^2 Dt}{L^2}\right) \text{ for } 0.4 \leq \frac{M_t}{M_\infty} \leq 1.0$$

- For release rate:

$$\text{EARLY TIME } \frac{dM_t}{dt} = 2M_\infty \left(\frac{D_{i,p}}{\pi L^2 t}\right)^{1/2}$$

$$\text{LATE TIME } \frac{dM_t}{dt} = \frac{8DM_\infty}{L^2} \exp\left(-\frac{\pi^2 Dt}{L^2}\right)$$



- A non-degradable polymer matrix slab was prepared with 100mg of drug dissolved in it. It release 5mg of drug on after 24 hours. How long will it take to release 25mg of drug?

$$\text{EARLY TIME } \frac{M_t}{M_\infty} = 4\left(\frac{Dt}{\pi L^2}\right)^{1/2} \text{ for } 0 \leq \frac{M_t}{M_\infty} \leq 0.6$$

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- A non-degradable polymer matrix slab was prepared with 100mg of drug dissolved in it. It release 5mg of drug on after 24 hours. How long will it take to release 25mg of drug?

$$\frac{5}{100} = 0.05$$

$$\frac{5}{100} = \left(4 \left(\frac{Dt}{\pi L^2} \right)^{1/2} \right) \frac{25 \text{mg}}{100} = 0.25$$

$$\frac{25}{100} =$$

$$t_n = 25t \approx 25 \text{ days} \propto \frac{1}{2}$$

EARLY TIME $\frac{M_t}{M_\infty} = 4 \left(\frac{Dt}{\pi L^2} \right)^{1/2}$ for $0 \leq \frac{M_t}{M_\infty} \leq 0.6$

LATE TIME $\frac{M_t}{M_\infty} = 1 - \frac{8}{\pi^2} \exp\left(-\frac{\pi^2 Dt}{L^2}\right)$ for $0.4 \leq \frac{M_t}{M_\infty} \leq 1.0$

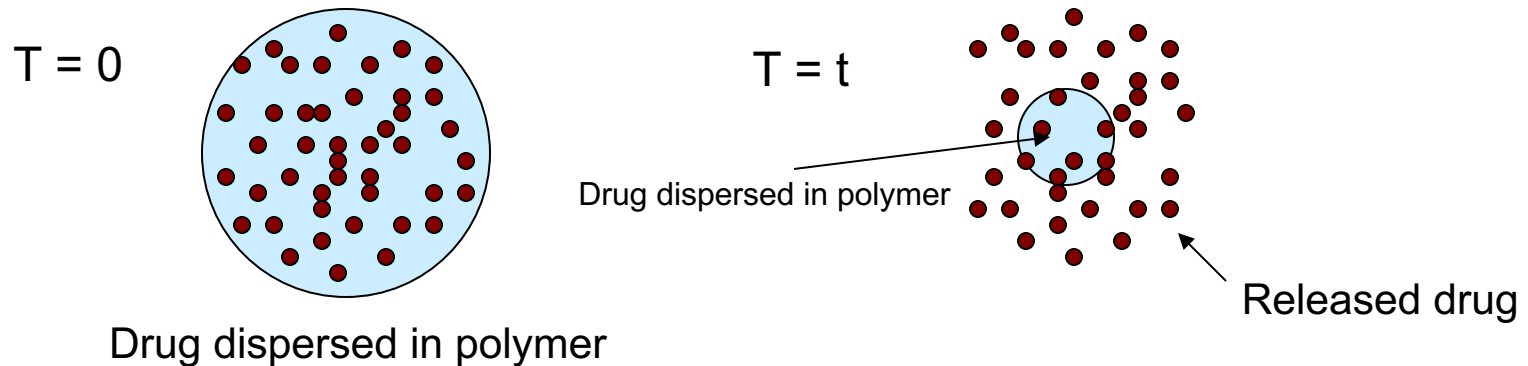
Matrix delivery: Example

- Compudose cattle growth implant
 - Microcrystalline estradiol dispersed in a silicone rubber matrix
 - Implant is placed under animal skin, estradiol is released and enters the systemic circulation, stimulates growth hormone production
 - Available to release estradiol over 200 or 400 days (by varying matrix thickness)
- EVAc vaginal implants for contraceptives
- EVAc implants for delivery of chemotherapy to the brain

Reservoir versus Matrix

- Reservoir systems
 - Zero-order release kinetics leading to constant drug release with time
 - More complicated fabrication process
- Matrix systems
 - Simple to fabricate
 - Drug release decreases with time
- Choice criteria
 - Physicochemical properties of the drug
 - Desired release rate and profile
 - Desired delivery duration
 - Ease of fabrication
 - Patient Safety

Bio-erodible systems



- **Advantages**

- Degradable, more patient compliance
- Can achieve zero order with surface eroding polymer devices

- **Disadvantages**

- Release kinetics is often difficult to control
- By-products of degradation may cause biocompatibility problems

Simplified models of erodible systems

- **Heterogeneous or Surface Erosion**

Simple slab (planar geometry) of area (of both sides) = A

Uniform drug concentration = C_0

- **For only erosion controlled release (no –diffusion)**, the release rate is proportional to surface erosion rate (which is assumed constant) and can be written as:

$$\frac{dM_t}{dt} = BC_0A, \quad \text{where } B = \text{surface erosion rate} \quad \begin{array}{l} B \text{ has units of} \\ 1/\text{hr.cm}^2 \end{array}$$

Integrating and assuming initial drug release = 0 (at $t = 0$), one gets

$$M_t = BC_0At$$

- The general form of release rate holds for different geometries and yields a solution with the general form

$$\frac{M_t}{M_\infty} = 1 - \left[1 - \frac{Bt}{C_0l}\right]^n$$

where $n = 1$ for slab of thickness $2l$, $n = 2$ for a cylinder of radius l and $n = 3$ for a sphere of radius l

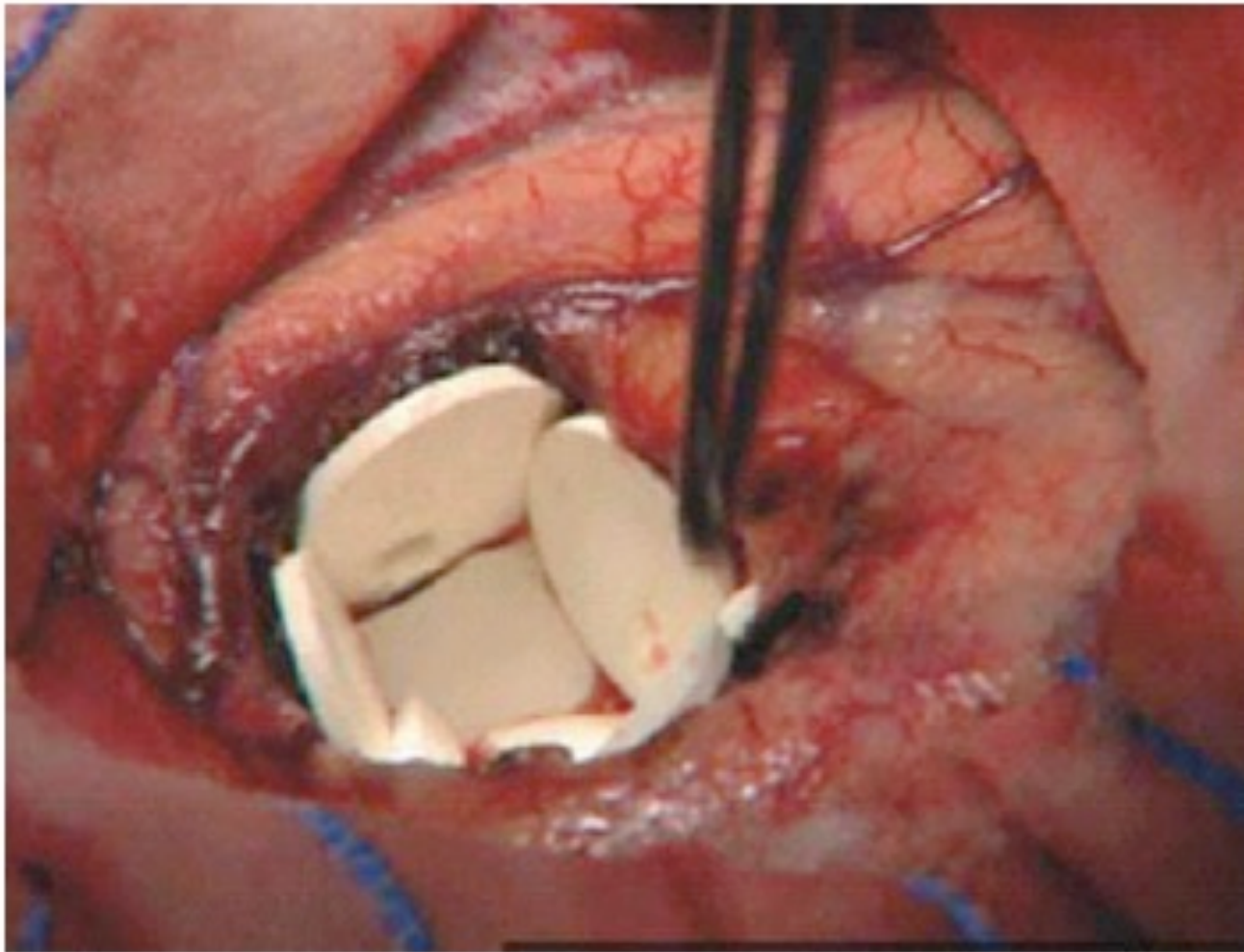
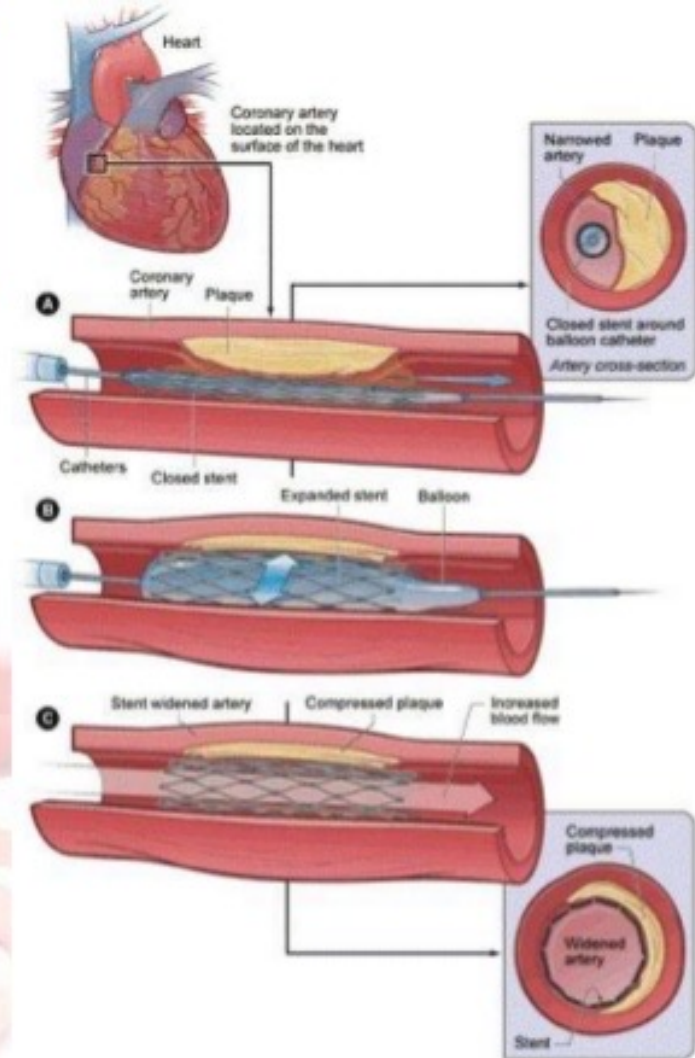


Fig. 4. Gliadel® wafer implantation at tumor resection site. Around 7–8 wafers are placed for localized delivery of BCNU via P(CPP:SA) polymer (Reproduced with permission from [22]).

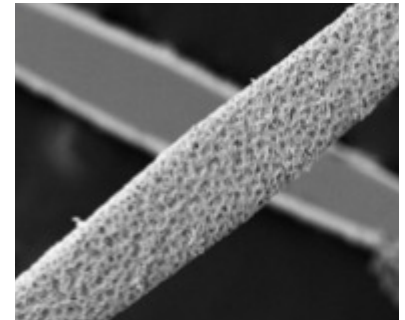
Copolymer of sebacic acid and carboxy phenoxy propane, and erodes over 6–8 weeks in vivo

Drug Eluting stents

- A metallic stent contains drug which is gradually released over 14-30 days
- Drugs used are :
Sirolimus, Paclitaxel, everolimus



- Coated with matrix containing drug to prevent muscle proliferation
 - Anti-coagulants are also used



Multi-pulse drug delivery from a resorbable polymeric microchip device

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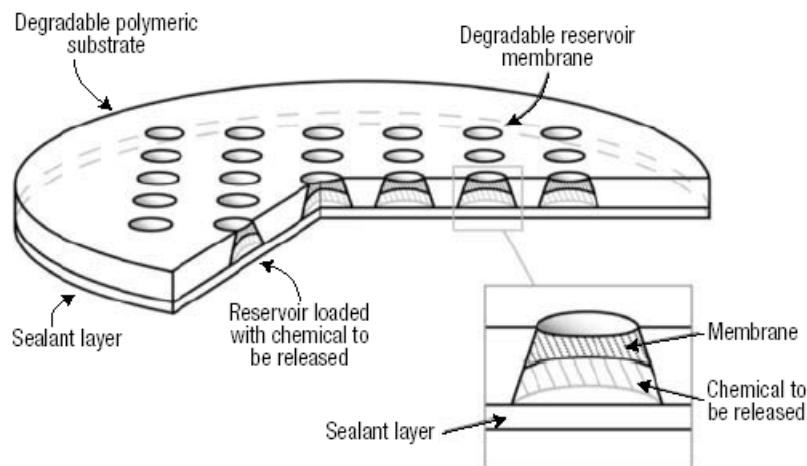
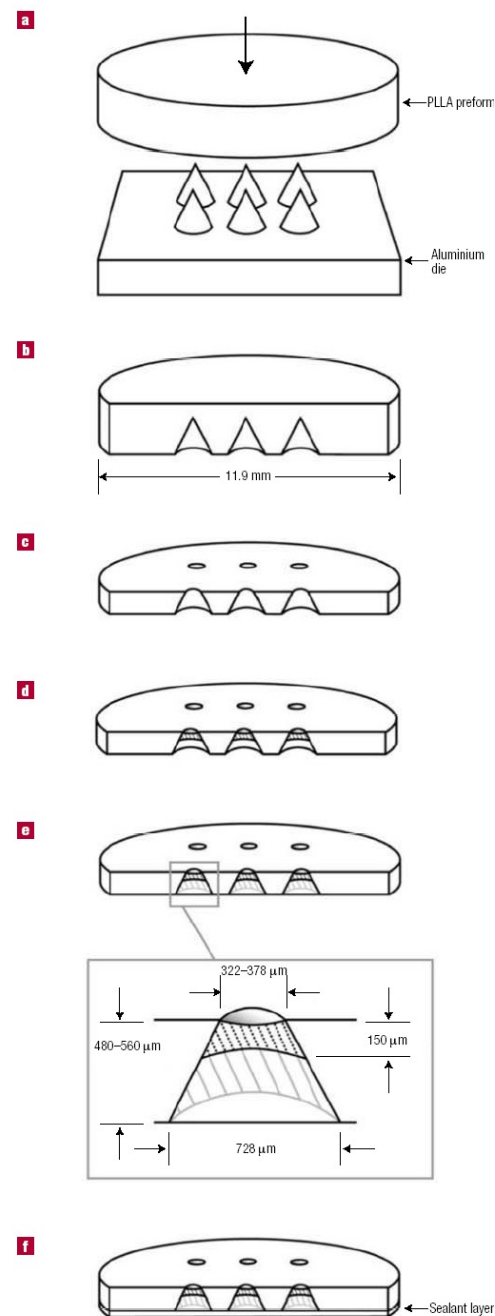


Figure 1 Diagram of polymeric microchip device. The main body of the device is composed of a reservoir-containing substrate that is fabricated from a degradable polymer. Truncated conical reservoirs in the substrate are loaded with the chemical to be released, and sealed with polymeric degradable reservoir membranes on one end and a sealant layer (polyester tape) on the opposite end. Inset, close-up of a reservoir, reservoir membrane, sealant layer and chemical to be released.



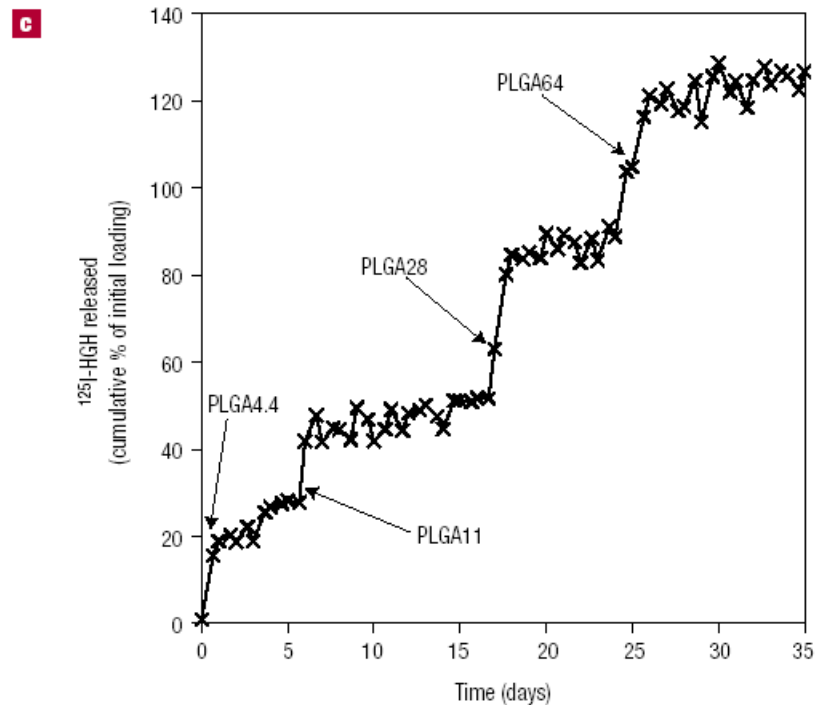
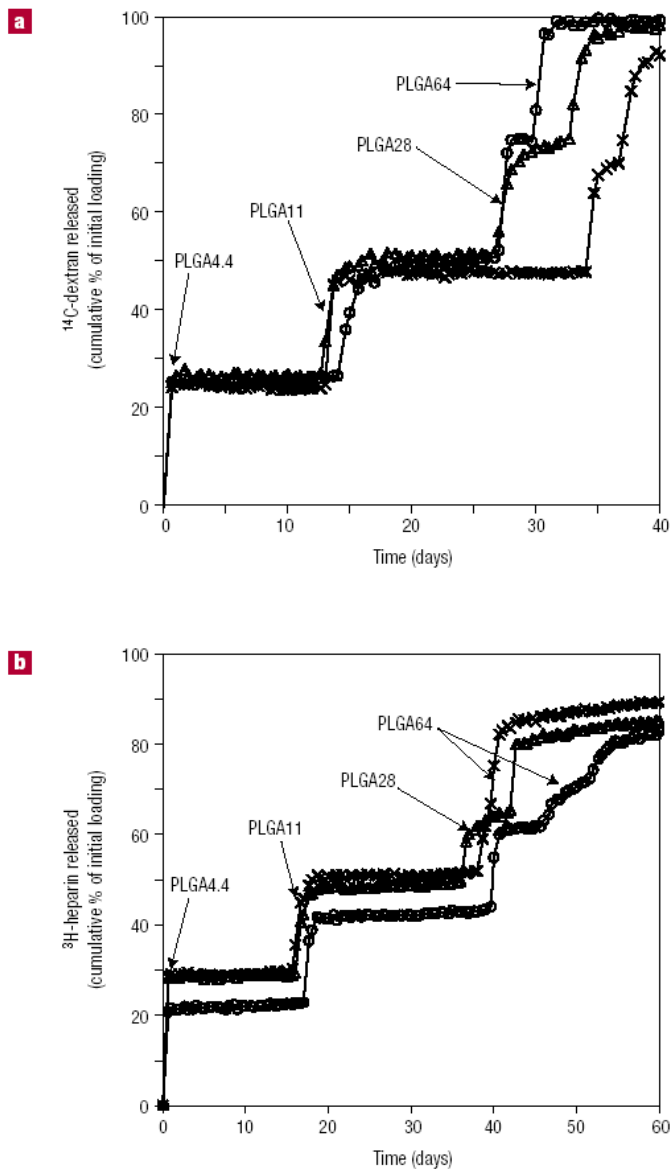


Figure 3 Cumulative percentage of initial loading released from microchip devices *in vitro*. Each symbol (triangle, cross or circle) in a panel represents data collected for a different device. Each device had a total of four reservoirs that were loaded with radiolabelled molecules and sealed with a membrane, and each reservoir had a different membrane that was fabricated from one of the PLGA4.4, PLGA11, PLGA28 or PLGA64 copolymers. The release times of the chemicals from the reservoirs increased as the molecular mass of the reservoir membrane polymers was increased, as shown by the arrows indicating the opening of each type of membrane on the devices. Experiments were conducted in saline solution at 28–33 °C *in vitro*. Devices represented by triangles are included for comparison⁴⁹. **a**, Cumulative release results for devices loaded with ¹⁴C-dextran. **b**, Cumulative release results for devices loaded with ³H-heparin. **c**, Cumulative release results for a device loaded with ¹²⁵I-HGH.

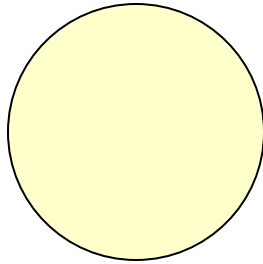
Hydrogels: What are they?

- Hydrogels are three dimensional, hydrophilic polymer networks capable of swelling by absorbing large amounts of fluid
 - Can expand to thousands of times their dry weight in fluid
- Insoluble due to physical or chemical crosslinks which provides the network structure and physical integrity
 - Could be bioerodible or non-erodible
- Numerous applications
 - Contact lenses, tissue engineering matrices, biosensor membranes, drug delivery etc.



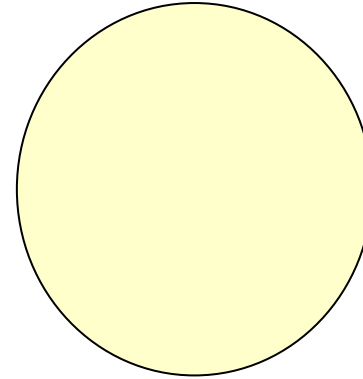
Swelling-controlled matrix (Hydrogels)

$T = 0$



Drug dissolved in polymer

$T = t$



- **Advantages**

- Low burst effects
- Reformulation of vehicle not necessary for different drugs

- **Disadvantages**

- Generally short release periods
- Not suitable for all delivery routes or targets

Hydrogels: Classifications

- **Physical hydrogels:**

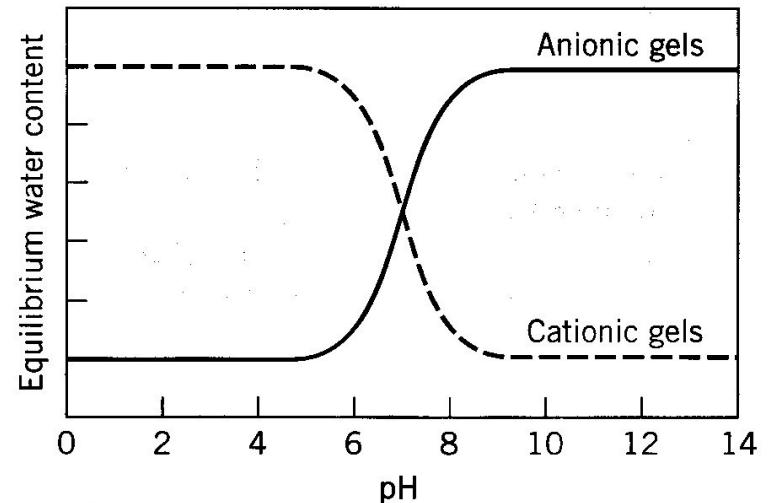
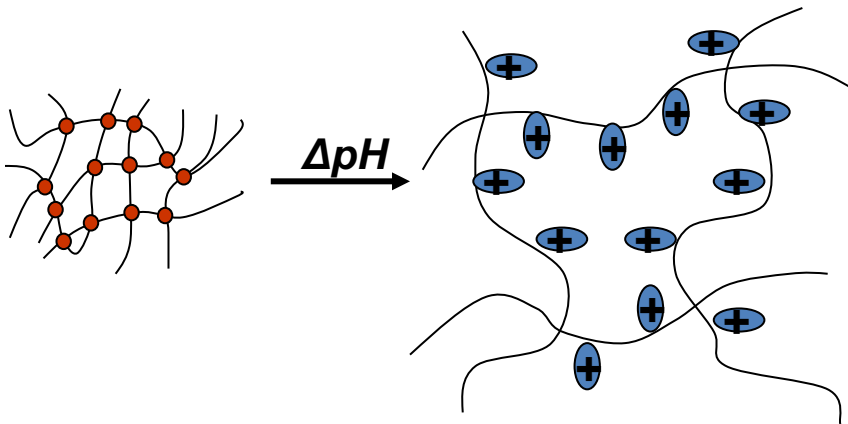
- The polymer networks are held together by neutral or ionic bonds
 - Molecular entanglements
 - Secondary forces including ionic, Hydrogen-bonding or hydrophobic forces
- **Not homogeneous** → clusters of molecular entanglements, or hydrophobically- or ionically-associated domains

Physical Hydrogels

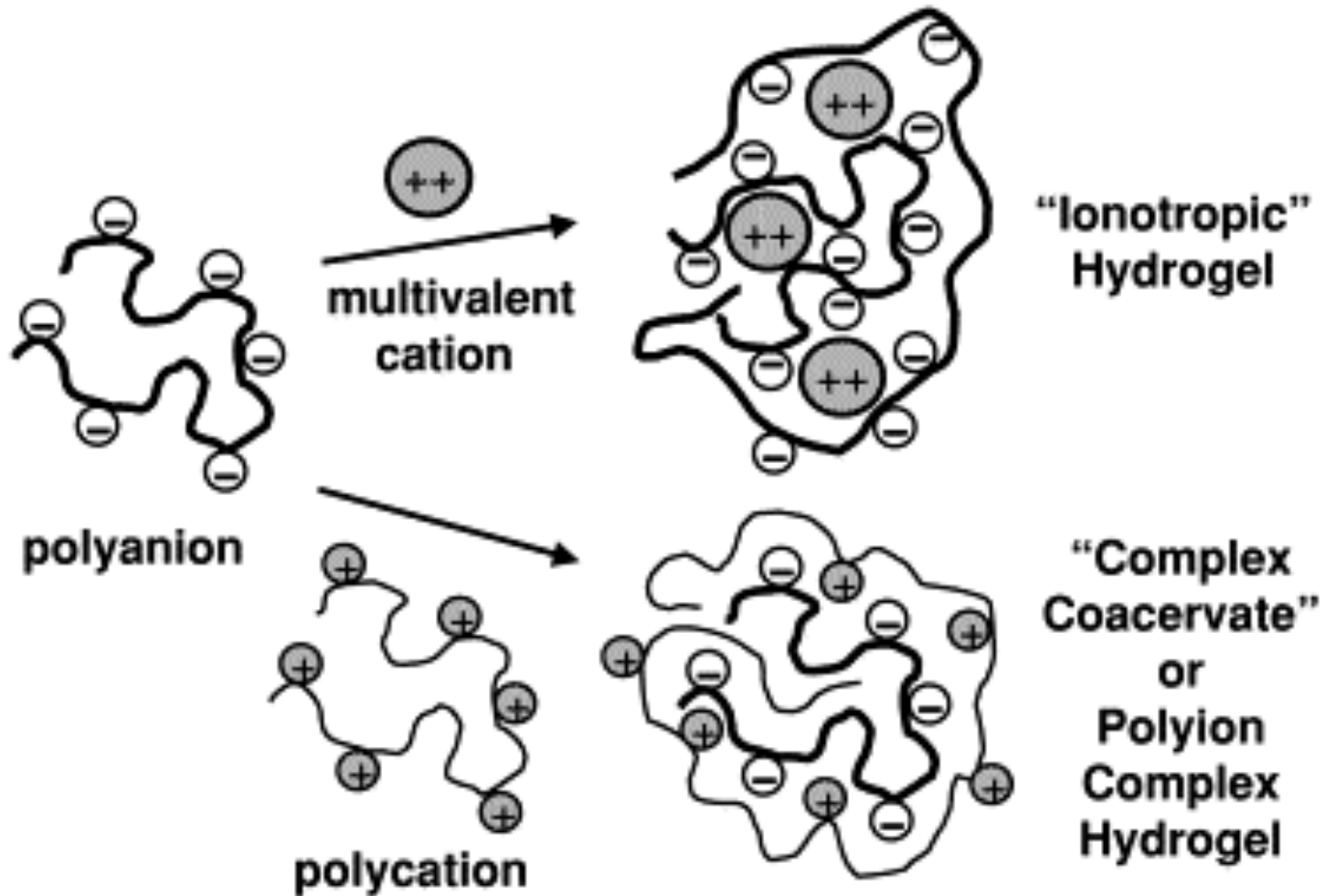
- Sometimes, physical gels can form from bio-specific recognitions,
 - Concanavalin A with a polymeric sugar or
 - Avidin with a polymeric biotin
- Reversible interactions → can be disrupted by physical factors
 - Ionic strength, pH, temperature, stress, or competing solutes
 - Can be used to release their cargo faster

Ionic Hydrogels

- Polymer chains within the hydrogel contains pendant groups that are cationic or anionic
- Ionic → Gels are unionized above (cationic) or below (anionic) the pKa of the groups. For pH above (anionic) or below (cationic) pKa, the pendant groups are ionized and repel each other → large swelling (completely reversible)



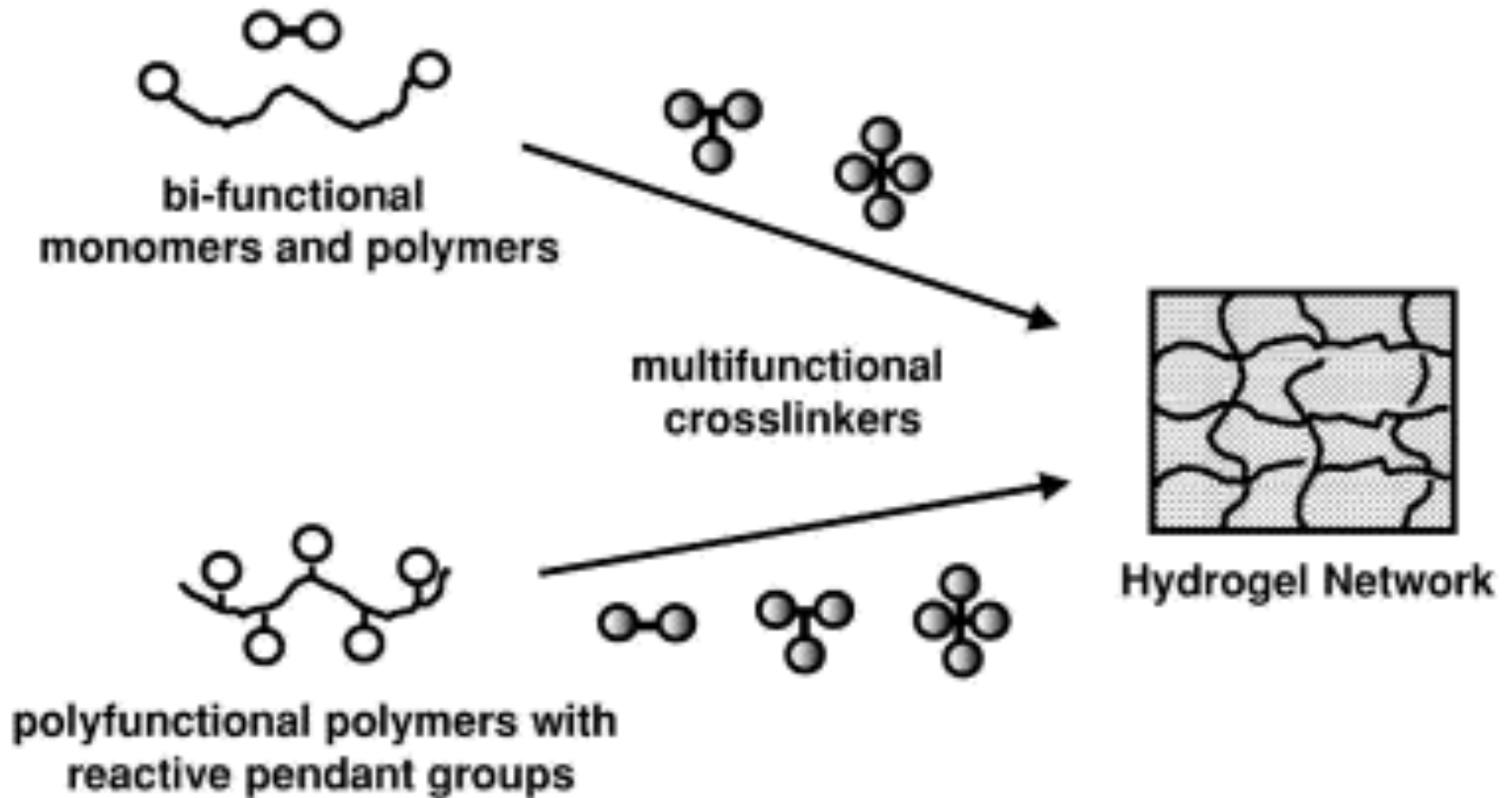
Ionic hydrogels



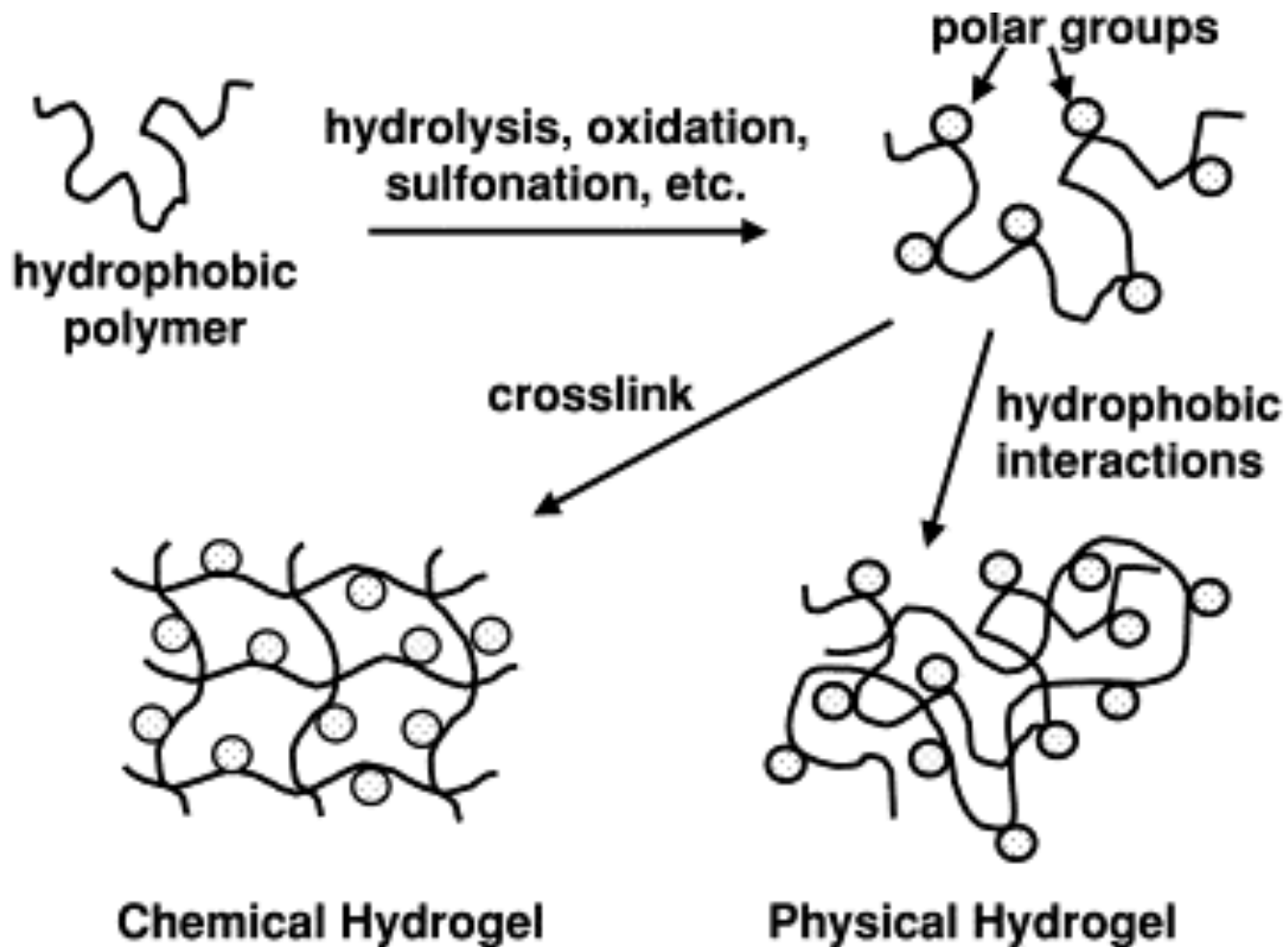
Chemical hydrogels

- Covalently-crosslinked hydrogel networks.
- May be generated by
 - directly crosslinking of water-soluble polymers, or by
 - conversion of hydrophobic polymers to hydrophilic polymers plus crosslinking to form a network.
- Like physical hydrogels, chemical hydrogels are not homogeneous.
 - Usually contain regions of low water swelling and high crosslink density, called 'clusters', that are dispersed within regions of high swelling, and low crosslink density. This may be due to hydrophobic aggregation of crosslinking agents, leading to high crosslink density clusters.

Hydrogel synthesis processes



Hydrogels from hydrophobic polymers

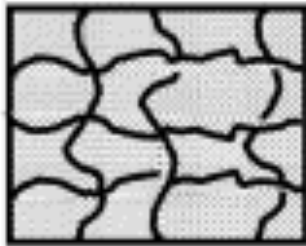


Interpenetrating network

water soluble polymer

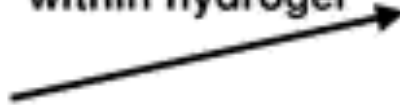


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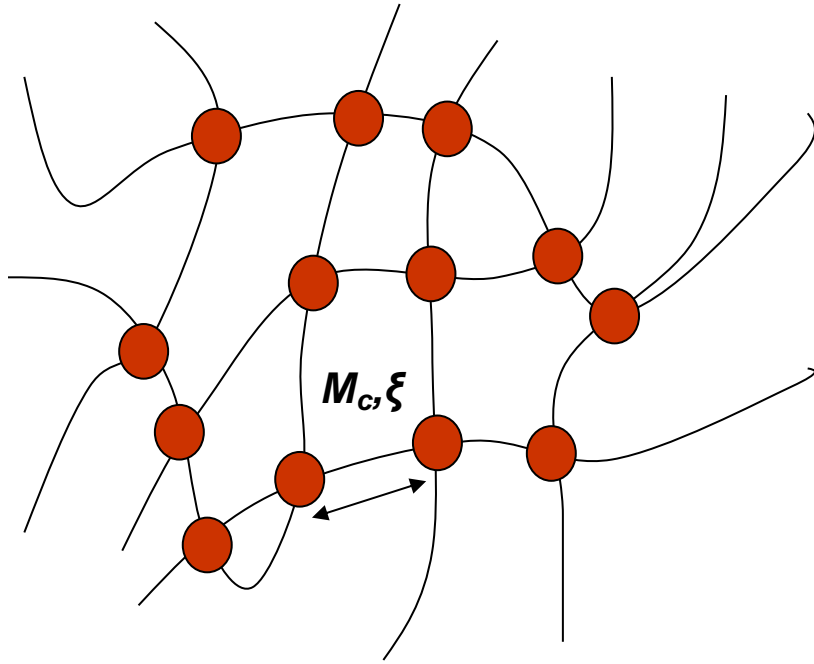
Hydrogel Network

polymerize monomer
within hydrogel



Interpenetrating
Hydrogel Network

Structure and properties of hydrogels



$$v_{2,s} = (\text{volume of polymer})/(\text{volume of swollen gel}) \\ = V_p / V_{gel} = 1/Q$$

- Most important parameters defining hydrogels
 - Polymer volume fraction in the swollen state ($v_{2,s}$)
 - A measure of the amount of fluid that a hydrogel can incorporate into its structure
 - Determined by equilibrium swelling experiments
 - Effective molecular weight of the polymer between crosslinks (M_c)
 - Measure of the degree of crosslinking in the gel
 - Network mesh or pore size (ξ)
 - Measure of the network porosity

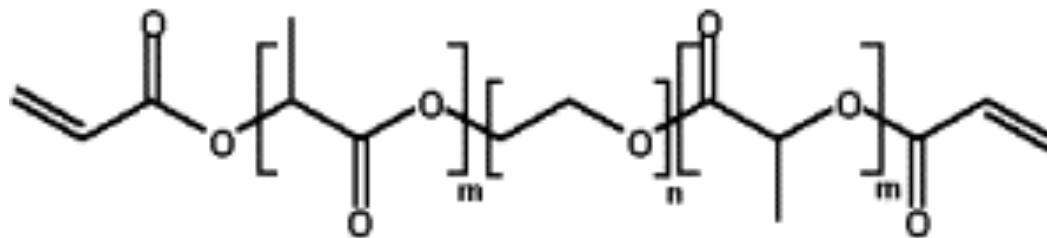
Neutral Hydrogels: Mathematical Treatment

- **Flory-Rehner theory for crosslinked polymers gels**
 - Based on **Gaussian distribution** of polymer chains
 - Degree to which a polymer network swelled, was governed by (a) the elastic retractive forces of the polymer chains and (b) the thermodynamic compatibility of the polymer and solvent molecules (mixing)
 - Therefore, the total free energy upon swelling is given by

$$\Delta G_{total} = \Delta G_{elastic} + \Delta G_{mixing}$$

Example: PLA-PEG-PLA hydrogels

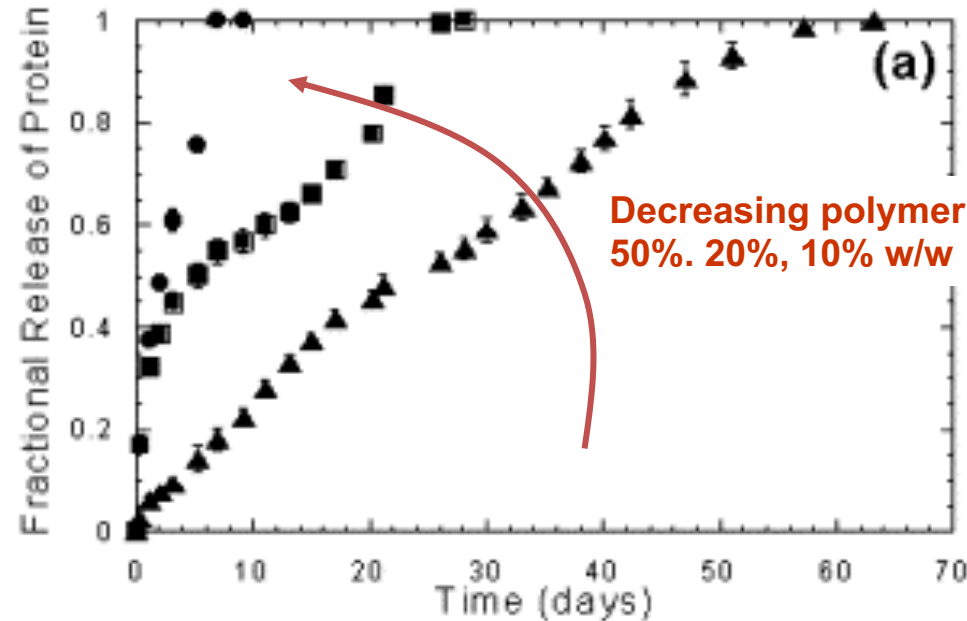
- PLA is polymerized to the hydroxyl end groups of PEG creating a block copolymer
- The terminal acid groups of PLA are acrylated and the polymer is crosslinked using UV light to form a hydrogel



Macromer 1: $m \sim 3, n \sim 104$

Macromer 2: $m \sim 2, n \sim 104$

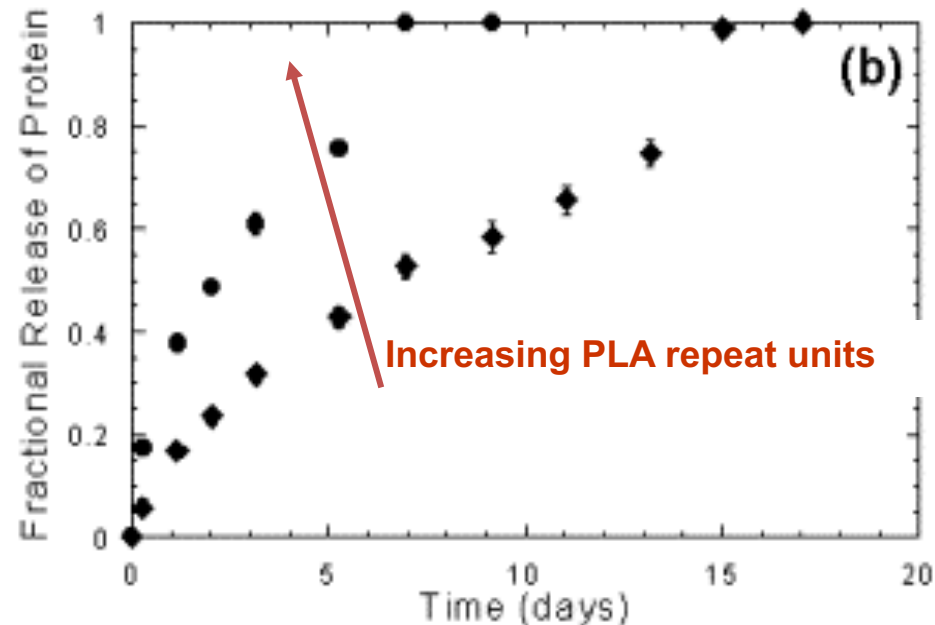
Release of BSA: Kinetics



Decreasing polymer concentration
50%, 20%, 10% w/w

Effect of decreasing
polymer concentration?

Effect of increasing the chain length of PLA
ends while keeping PEG length same?



Increasing PLA repeat units

In situ Crosslinking Hydrogels

- Cross-linked at the site – physician has a syringe and injects into body
- Forms a hydrogel inside the body
- Polymerization trigger could be:
 - Time
 - Temperature
 - pH
- Disadvantages
 - May diffuse throughout body
 - No control of shape

- Currently there are over 30 injectable hydrogel-based products that have been approved by the FDA and/or EMA (Table 1)

TABLE 1 Clinically approved injectable hydrogels, grouped by their material class and broad indication

Name (company)	Hydrogel material/payload (gelation mechanism)	Injection type	Approved indication	Approval (year)
Cancer: synthetic				
SpaceOAR [®] Hydrogel (Augmenix, Inc.)	Polyethylene glycol (chemical reaction)	Percutaneous	For protecting vulnerable tissues during prostate cancer radiotherapy	EMA (2010) FDA (2015)
Vantas [®] (Endo Pharmaceuticals)	Histrelin acetate, poly(2-hydroxyethyl methacrylate), poly(2-hydroxypropyl methacrylate) and gonadotropin releasing hormone (chemical reaction)	Subcutaneous	Palliative treatment of prostate cancer	FDA (2004) EMA (2005)
Facial correction: synthetic				
Radiesse [®] (+) (Merz Pharmaceuticals)	Hydroxylapatite, carboxymethylcellulose with Lidocaine (physical interaction)	Dermis	Correction of wrinkles and folds, stimulation of natural collagen production	FDA (2015)
Radiesse [®] (Bioform Medical, Inc.)	Hydroxylapatite, carboxymethylcellulose (physical interaction)	Dermis	For correction of facial folds and wrinkles, signs of facial fat loss and volume loss	EMA (2004) FDA (2006 for first indication)
Artefill [®] (Suneva Medical, Inc.)	Polymethylmethacrylate beads, collagen and lidocaine (physical interaction)	Dermis	Facial wrinkles and folds	FDA (2006)
Sculptra [®] (Sanofi Aventis U.S.)	Poly-L-lactic acid (physical interaction)	Dermis	For correction of signs of facial fat loss, shallow to deep contour deficiencies and facial wrinkles	EMA (2000) FDA (2004 for first indication)

Facial correction: natural

Belotero balance® (+) Lidocaine (Merz Pharmaceuticals)	Hyaluronic acid with lidocaine (chemical reaction)	Dermis	Moderate to severe facial wrinkles and folds	FDA (2019)
Revanesse® Versa+	Hyaluronic acid with lidocaine (chemical reaction)	Dermis	Moderate to severe facial wrinkles and creases	FDA (2018)
Teosyal® RHA (Teoxane SA)	Hyaluronic acid (chemical reaction)	Dermis	Facial wrinkles and folds	EMA (2015) FDA (2017)
Revanesse® Versa/Revanesse® Ultra (Prolenium Medical Technologies Inc.)	Hyaluronic acid (chemical reaction)	Dermis	Moderate to severe facial wrinkles and creases	FDA (2017)
Restylane® Lyft, Restylane® Refyne, Restylane® Defyne (Galderma Laboratories, L.P.)	Hyaluronic acid with Lidocaine (chemical reaction)	Subcutaneous, dermis, lips	For correction of volume deficit, facial folds and wrinkles, midface contour deficiencies, and perioral rhytids	EMA (2010) FDA (2012 for first indication)
Restylane® Silk (Valeant Pharmaceuticals North America LLC/Medicis)				
Restylane® Injectable Gel (Medicis Aesthetics Holdings, Inc.)				
Belotero balance® (Merz Pharmaceuticals)	Hyaluronic acid (chemical reaction)	Dermis	Moderate to severe facial wrinkles and folds	EMA (2004) FDA (2011)
Juvederm® XC (Allergan, Inc.)	Hyaluronic acid with lidocaine (chemical reaction)	Facial tissue	Correction of facial wrinkles and folds	FDA (2010)
Evolence® Collagen Filler (Colbar Lifescience I)	Collagen (chemical reaction)	Dermis	Moderate to deep facial wrinkles and folds	EMA (2004) FDA (2008)
Eleess® (Anika Therapeutics)	Hyaluronic acid with lidocaine (chemical reaction)	Dermis	Moderate to severe facial wrinkles and folds	FDA (2006) EMA (2007)

TABLE 1 (Continued)

Name (company)	Hydrogel material/payload (gelation mechanism)	Injection type	Approved indication	Approval (year)
Juvéderm [®] /Voluma XC/Ultra XC/Volbella XC/ Vollure XC (Allergan, Inc)	Hyaluronic acid (chemical reaction)	Facial tissue, cheek, lips	For correction of facial wrinkles and folds, volume loss, and lip augmentation.	EMA (2000) FDA (2006 for first indication)
Hylaform [®] (Hylan B gel), Captique Injectable Gel, Prevelle Silk (Genzyme Biosurgery)	Modified hyaluronic acid derived from a bird (avian) source (chemical reaction)	Dermis	Correction of moderate to severe facial wrinkles and folds	EMA (1995) FDA (2004)
Collagen Implant, CosmoDerm [®] 1 human-based collagen, CosmoDerm [®] 2 human-based collagen CosmoPlast [®] human-based collagen (Inamed Corporation/Allergan, Inc.)	Human collagen (CosmoDerm: physical interaction, CosmoPlast: chemical reaction)	Superficial papillary dermis	For correction of soft tissue contour deficiencies, such as wrinkles and acne scars	FDA & EMA (2003)
Fibrel [®] (Serono Laboratories)	Collagen (physical interaction)	Dermis	For correction of depressed cutaneous scars	FDA (1988)
Zyplast(R) [®] and Zyderm(R) [®] (Inamed Corporation/Allergan, Inc.)	Bovine collagen (chemical reaction)	Dermis	For correction of contour deficiencies	FDA and EMA (1981)
Spinal fusion: natural				
EUFLEXA [®] (Ferring Pharmaceuticals Inc.)	Hyaluronic acid (physical interaction)	Intra-articular	Knee osteoarthritis	FDA (2004) EMA (2005)
INFUSE [®] bone graft (Medtronic Sofamor Danek USA, Inc.)	Collagen and recombinant human bone morphogenetic protein-2 (physical interaction)	Spinal injection	Spinal fusion, and spine, oral-maxillofacial and orthopedic trauma surgeries	FDA (2002 for first indication)

Osteogenic protein 1(OP-1®) implant, OP-1® Putty (Stryker Biotech)	Collagen, carboxymethylcellulose, and recombinant OP-1 (physical interaction)	Spinal injection	Posterolateral lumbar spinal fusion	FDA (2001)
Other: synthetic				
TracelT® Hydrogel Tissue Marker (Augmenix, Inc.)	Polyethylene glycol (chemical reaction)	Percutaneous	Improved soft tissue alignment for image guided therapy	FDA (2013)
Supprelin LA® (Indevus Pharmaceuticals, Inc.)	Histrelin acetate, Poly (2-hydroxyethyl methacrylate) (chemical reaction)	Subcutaneous	Central precocious puberty	EMA (2005) FDA (2007)
Bulkamid® hydrogel (Searchlight Pharma)	Polyacrylamide (chemical reaction)	Transurethral	Female stress urinary incontinence	EMA (2003) FDA (2006)
Coaptite® (BioForm Medical, Inc.)	Calcium hydroxylapatite, sodium carboxymethylcellulose, glycerin (physical interaction)	Submucosal	Female stress urinary incontinence	EMA (2001) FDA (2005)
Other: natural				
Algisyl-LVR® Hydrogel Implant (LoneStar Heart, Inc.)	Alginate (physical interaction)	Percutaneous	Advanced heart failure	EMA (2014)